

- 5 A molecule of collagen consists of three identical polypeptides that form a triple helix.

The amino acid glycine forms one third of the amino acids in a collagen molecule.

Fig. 5.1**A** shows a polypeptide molecule during protein synthesis. A molecule of glycine is shown just before it is added to the polypeptide.

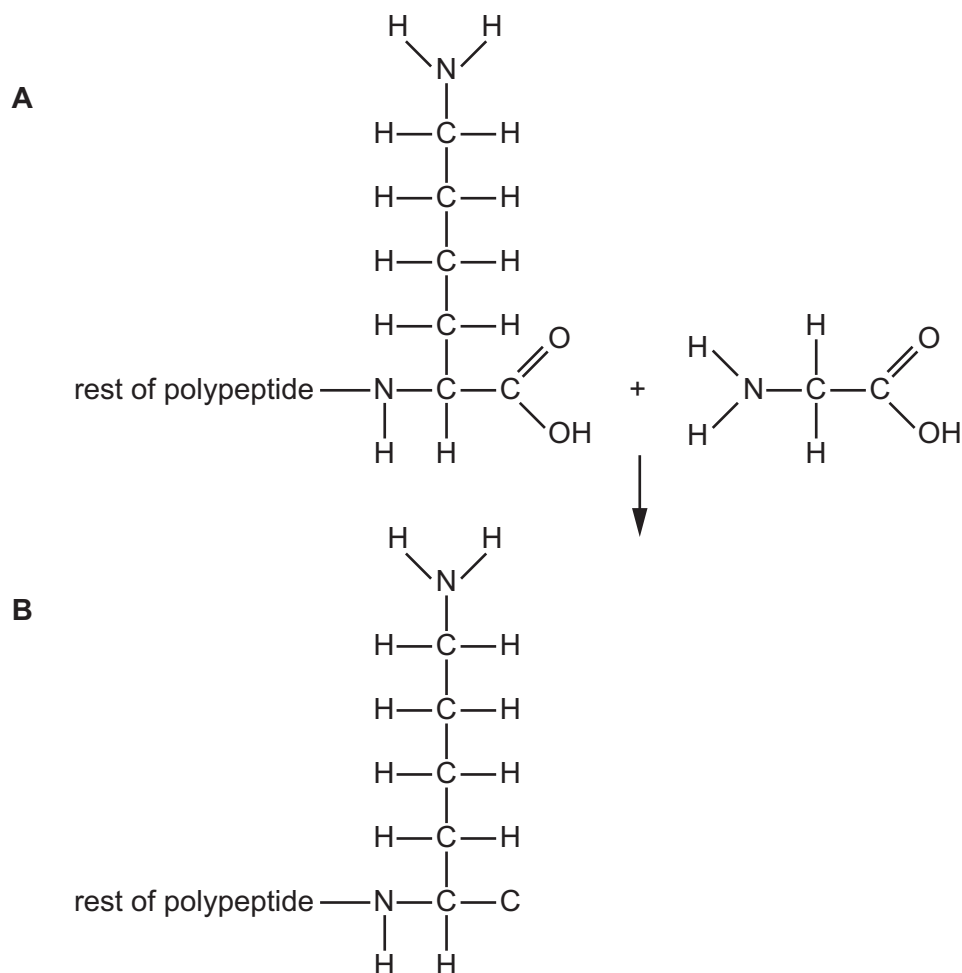


Fig. 5.1

- (a) (i) Complete Fig. 5.1**B** to show the molecule of glycine added to the end of the polypeptide. [2]

- (ii) State the type of reaction that occurs when glycine is added to the end of the polypeptide.

..... [1]

- (iii) Explain the importance of glycine in a collagen molecule.

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..... [2]

- (b)** Collagen is a fibrous protein.

Explain how the structure of collagen is related to its functions within the mammalian body.

[4]

- (c)** The enzyme collagenase catalyses the breakdown of collagen molecules. This enzyme acts by using an induced fit mechanism.

Describe the mode of action of collagenase.

[5]

- (d) Students investigated the effect of pH on the activity of collagenase extracted from the bacterium *Bacillus licheniformis* at 50 °C.

The results are shown in Fig. 5.2.

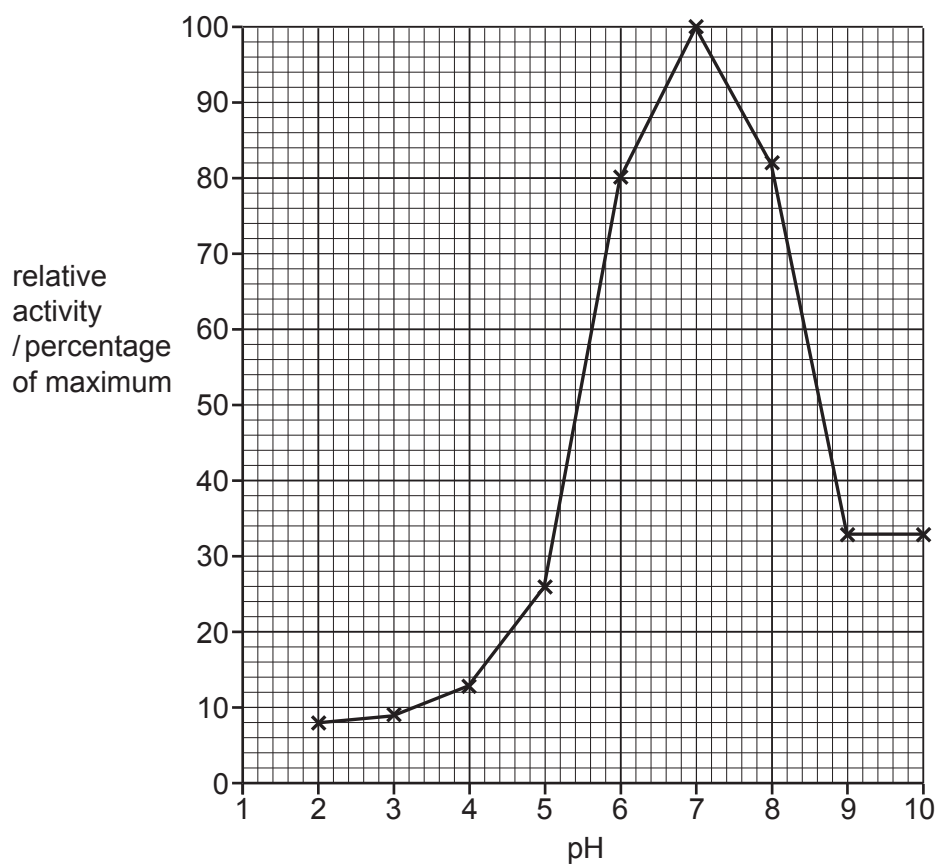


Fig. 5.2

With reference to Fig. 5.2, describe **and** explain the effect of pH on the activity of collagenase.

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[Total: 18]